TEACHER GUIDE · GRADE 2 · BLOCK CODING

Debug the Pair

Quick facilitation notes & answer key

Goal

The challenge sheet: students DEBUG a pair of concurrent scripts (C3.2 read/alter + C3.1 write). The two sprites are meant to meet on 3, but Pixel's script lands it on 4. Students check each script against the goal, identify Pixel's as the buggy one (because Bit already reaches 3), and write the fix (back 1 → back 2) so the pair meets.

Ontario curriculum (Grade 2 — C3 Coding)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential and concurrent events.

C3.2 — read and alter existing code, including code that involves sequential and concurrent events, and describe how changes to the code affect the outcomes.

Materials

The printed worksheet and a pencil. Optional: two counters on a 0-5 path and a star on square 3 (the meeting goal) so students can see which sprite misses the target.

Run it (about 20 min)

1. I DO: state the goal — 'both should meet on 3.' Run both: Bit → 3 (on target), Pixel → 4 (misses).
2. WE DO: reason about WHICH is wrong — since Bit already lands on 3, the bug must be in Pixel's script.
3. YOU DO: students write Pixel's fix (back 2), run again, and confirm both land on 3 — debugged!
4. Connect: 'To debug two scripts, check each one against the goal — the one that misses is the bug.'

Answer key (modelled by the run-gate)

Ex 1 — Bit: 0 → forward 3 → 3. Pixel: 5 → back 1 → 4. They do NOT meet on 3 (Pixel is on 4).

Ex 2 — Pixel's script is the wrong one. Bit already lands on the target 3, so the bug must be in Pixel's.

Ex 3 — fix Pixel to 'back 2': 5 → back 2 → 3. Now both land on 3 → they MEET. (Accept any Pixel script that lands on 3, e.g. 'back 1, back 1'.)

Interaction note: only the buggy script needs changing; once Pixel reaches 3, the two concurrent scripts meet on the target.

Watch for & differentiate

Common error: changing Bit's script — Bit already lands on 3, so changing it breaks a correct script. Stress checking each against the goal first.

Common error: fixing the landing but not re-checking 'do they meet?' — have students confirm BOTH are on 3 after the fix.

Simplify: tell students Bit is correct and only Pixel needs fixing; have them just find Pixel's move to 3.

Extend: change the goal to 'meet on 2' and ask which script(s) need fixing now (both).

Success indicator: the child names Pixel's script as the bug and writes a fix that lands Pixel on 3 so the pair meets.